

Pulkit Verma

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Research Interests

AI Safety, AI Planning, Action-Model Learning, Analysis of Abstractions, and Robotics.

Education

Ph.D. in Computer Science, Arizona State University

Advisor: Prof. Siddharth Srivastava | GPA: 4.0/4.0

Tempe, USA

Exp. Fall 2023

Master of Technology in Computer Science & Engineering, IIT Guwahati

Advisor: Prof. P. K. Das | Thesis: Resource Usage Analysis for Speech Recognition Techniques | GPA: 9.07/10

Guwahati, India

May 2015

Bachelor of Engineering in Information Technology, SVITS, Indore

Advisor: Prof. Anand Rajawat | Capstone: Smart Analyzer - Predict and Cache Webpages based on Usage.

Indore, India

June 2011

Publications

Journals and Conferences

Pulkit Verma, Shashank Rao Marpally, and Siddharth Srivastava. "Discovering User-Interpretable Capabilities of Black-Box Planning Agents". In *19th Intl. Conference on Principles of Knowledge Representation and Reasoning*, 2022. 📄

Rashmeet Kaur Nayyar*, **Pulkit Verma***, and Siddharth Srivastava. "Differential Assessment of Black-Box AI Agents". In *36th AAAI Conference on Artificial Intelligence*, 2022. 📄 🎥 *Equal Contribution

Yizhong Wang et al. "Super-NaturalInstructions: Generalization via Declarative Instructions on 1600+ Tasks". In *2022 Conference on Empirical Methods in Natural Language Processing*, 2022. 📄

Pulkit Verma, Shashank Rao Marpally, and Siddharth Srivastava. "Asking the Right Questions: Learning Interpretable Action Models Through Query Answering". In *35th AAAI Conference on Artificial Intelligence*, 2021. 📄 🎥

Pulkit Verma, and Pradip K. Das. "A Comparative Study of Resource Usage for Speaker Recognition Techniques". In *International Conference on Signal Processing and Communication*, 2016. 📄

Pulkit Verma, and Pradip K. Das. "i-Vectors in Speech Processing Applications: A Survey". In *International Journal of Speech Technology*, 18(4), pp.529-546, 2015. 📄

Mayank Gupta, **Pulkit Verma**, Tuhin Bhattacharya, and Pradip K. Das. "A Mobile Agents based Distributed Speech Recognition Engine for Controlling Multiple Robots". In *International Conference on Advances In Robotics*, 2015. 📄

Pulkit Verma, Mayank Gupta, Tuhin Bhattacharya, and Pradip K. Das. "Improving Services using Mobile Agents-based IoT in a Smart City". In *2014 International Conference on Contemporary Computing and Informatics*, 2014. 📄

Workshops, Demos, Preprints, DCs

Naman Shah*, **Pulkit Verma***, Trevor Angle, and Siddharth Srivastava. "JEDAI: A System for Skill-Aligned Explainable Robot Planning". In *21st International Conference on Autonomous Agents and MultiAgent Systems (Demonstration Track)*, 2022. [Best Demo Award]. 📄 🎥 *Equal Contribution

Pulkit Verma. "Data Efficient Paradigms for Personalized Assessment of Taskable AI Systems". In *ICAPS 2022 Doctoral Consortium*, and *KR 2022 Doctoral Consortium*, 2022. 📄

Pulkit Verma, and Siddharth Srivastava. "Learning Causal Models of Autonomous Agents using Interventions". In *IJCAI 2021 Workshop on Generalization in Planning*, 2021. 📄

Alok Shankar Mysore et al. "Investigating the 'Wisdom of Crowds' at Scale.". In *Adjunct Proceedings of the 28th Annual ACM Symposium on User Interface Software & Technology (Poster)*, 2015. 📄

Pulkit Verma, Shashank Rao Marpally, and Siddharth Srivastava. "Learning User-Interpretable Descriptions of Black-Box AI System Capabilities". In *ICAPS 2021 Workshop on Knowledge Engineering for Planning & Scheduling*, 2021. 📄

Pulkit Verma, and Siddharth Srivastava. "Learning Generalized Models by Interrogating Black-Box Autonomous Agents". In *AAI 2020 Workshop on Generalization in Planning*, 2020. 📄

Professional Experience

Arizona State University

Graduate Research Associate

Tempe, USA

Aug. 2018 - (present)

- Investigating the minimal set of requirements in an AI system that would enable a user to assess and understand the limits of its safe operability.
- Developed a framework which generates interrogation policies to derive an interpretable model of an AI agent.
- Co-lead a team to create an educational tool “JEDAI” to introduce AI planning concepts to laypersons using mobile manipulator robots.
- Co-designed the experiments showcasing possible use-cases of an NSF project on training lay people to use adaptive AI systems which are safe, robust and reliable.

NetApp Inc.

Storage Efficiency Developer

Bengaluru, India

Jul. 2015 - Jul. 2018

- *Workload Prediction*: Predicted workload on a storage volume based on I/O patterns for All Flash systems. This helped in optimizing data storage and management while keeping the nature of data private to the users.
- *File and Volume Clones (FVCs)*: Developed features and managed issues related to file and volume clones on UNIX-based NetApp proprietary WAFL file system to improve the storage efficiency of enterprise data servers.
- Co-developed a file system component which prevented data loss if FVCs are modified when cloud backup is offline.

Tata Consultancy Services Ltd.

iOS Application Developer

Pune, India

Mar. 2012 - Aug. 2013

- Built interactive applications for iPhone and iPad and developed tools to analyze the user investment portfolio.
- Performed cross-platform application development of mobile applications supported on both Android and iOS.

Teaching Experience

CSE 571 Artificial Intelligence [Graduate-level Course]

Arizona State University

Tempe, USA

Spring'22, Fall'19

- Delivered hands-on sessions on Planning Graphs, Markov Decision Processes, Reinforcement Learning, and Neural Networks in Spring 2022. *[Won ASU GPSA Teaching Excellence Award in Spring 2022 for this].*
- Delivered tutorials and readings on Robot Operating System (ROS), Constraint Satisfaction Problems (CSPs), First Order Logic, Planning Domain Definition Language (PDDL), and Dynamic Bayesian Networks in Fall 2019.
- Co-created the software test-bed used to support the ROS-based course projects.
- Created and graded homework assignments and exams.
- Moderated research paper discussions in lectures.
- Mentored project groups working on course projects.

Session Lead [Graduate-level Courses]

Arizona State University

Tempe, USA

2020-2022

- Delivered tutorials on *Reinforcement Learning* and *Neural Networks* in CSE 471 Intro. to AI, Fall'22 (85 students).
- Moderated three research paper discussions in CSE 574 Planning and Learning Methods in AI, Spring'21 (39 students).
- Delivered tutorial on *Probabilistic Inference* in CSE 471 Intro. to AI, Fall'20 (118 students).

CS 566 Speech Processing [Graduate-level Course]

Indian Institute of Technology Guwahati

Guwahati, India

Fall 2014

- Co-developed a custom C++ API to use a voice recording tool that students used in course projects.
- Mentored students working on course projects and homework assignments.
- Graded homework assignments and exams.

Undergraduate Lab. Courses

Indian Institute of Technology Guwahati

Guwahati, India

2013-2015

- TA for the lab courses: CS 513 Programming Lab.; CS 462 Graphics Lab.; and CS 243 Software Engineering Lab.
- Mentored project groups working on course projects.
- Graded homework assignments and exams.

Service

Co-organizer	IJCAI-ECAI 2022 Workshop on Generalization in Planning (GenPlan'22) ↗
PC Member	NeurIPS 2023 ↗ , IJCAI 2023 ↗ , AAMAS 2023 ↗ , AAI 2023 ↗ , HAXP 2023 ↗ , ODRUM 2023 ↗ , ICAPS 2022 ↗ , XAIP 2022 ↗ , XAIP 2021 ↗ , GenPlan 2021 ↗
Reviewer	IEEE Robotics and Automation Letters ↗ , Graduate and Professional Student Association, ASU ↗
Volunteer	IJCAI-ECAI 2022 ↗ , KR 2022 ↗ , IJCAI 2021 ↗ , ICAPS 2021 ↗ , ICLR 2021 ↗ , KR 2018 ↗ , Southwest Robotics Symposium 2019 ↗ , Workshop on GPU Programming and Applications 2014, IITG ↗

Honors & Awards

January 2023	SCAI Doctoral Fellowship , School of Computing and AI, Arizona State University
July 2022	IJCAI-AIJ 2022 Travel and Accessibility Grant , IJCAI-ECAI 2022
July 2022	KR 2022 Student Grant , KR 2022
May 2022	Best Demo Award , AAMAS 2022
May 2022	ICAPS Student Scholarship , ICAPS 2022
April 2022	GPSA Teaching Excellence Award , Arizona State University
March 2022	SCAI Doctoral Fellowship , School of Computing and AI, Arizona State University
2021-2022	GPSA Travel Grant , Arizona State University (for AAI 2021, AAI 2022, IJCAI 2022, KR 2022)
Mar 2021	Graduate College Travel Awards , ASU (for AAMAS 2021, ICLR 2021, ICML 2021, IJCAI 2021)
Oct 2018	Fulton Schools Experiential Learning Grant , Arizona State University
Aug 2013 - May 2015	Post Graduation Fellowship , All India Council for Technical Education, Government of India
June 2012	TCS ILP Kudos Award , Tata Consultancy Services Ltd. (Best Trainee in the new-joiner batch)

Mentorship

Kyle J. Atkinson

M.S. in Computer Science, Arizona State University

Tempe, USA

2021-2023

- Mentored Kyle at ASU where he worked on model learning for black-box AI systems in stochastic environments.
- Resulted in one submitted paper which is still under review.

Trevor Angle

M.S. in Computer Science, Arizona State University

Tempe, USA

2021-2022

- Co-mentored Trevor at ASU where he worked on a system for skill-aligned explainable robot planning, called JEDAI.
- Resulted in one short paper published in demonstration track at AAMAS 2022.

Shashank Rao Marpally

M.S. in Robotics and Autonomous Systems, Arizona State University

Tempe, USA

2020-2021

- Mentored Shashank for 3 semesters during his masters where he worked on learning action models for game agents.
- Resulted in two conference publications (AAAI'21 and KR'22).

Masih Tamsoy

B.Tech. in Computer Science and Engineering, IIT Guwahati

Guwahati, India

2014-2015

- Mentored Masih for 2 semesters at Robotics Lab, IIT Guwahati where he worked on his bachelor's thesis titled "Speaker Verification Systems".
- Helped Masih in developing a speaker verification system using joint factor analysis and i-vectors.

Other Affiliations

Jan 2021 - (present)	Graduate Student , Center for Human, Artificial Intelligence, and Robot Teaming, ASU ↗
Nov 2020 - (present)	Affiliated Graduate Student , Center for Human-Compatible Artificial Intelligence, UCB ↗
Nov 2020 - (present)	Member , Association for the Advancement of Artificial Intelligence (AAAI) ↗
July 2019	Participant , 3rd Summer School on Cognitive Robotics, USC ↗
Aug 2018 - (present)	Graduate Student , Autonomous Agents and Intelligent Robots (AAIR) Lab, ASU ↗
Jul 2016 - Dec 2016	Online Contributor , Stanford Scholar Initiative ↗
Jan 2014 - May 2015	Graduate Student , Robotics Lab, IIT Guwahati ↗